

Project 2: YSU Quantitative Finance

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June - 17, 2026

Outline

Bayesian Updates: Bernoulli Data

Model each binary outcome $X_i \in \{0, 1\}$ (e.g. is the daily return positive?) as $X_i \stackrel{iid}{\sim} \text{Bernoulli}(p)$.

Prior: $p \sim \text{Beta}(a, b)$, default $a = b = 1$ (Uniform)

Posterior (conjugacy), with $s = \sum x_i$ successes out of n :

$$p \mid \text{data} \sim \text{Beta}(a + s, b + n - s)$$

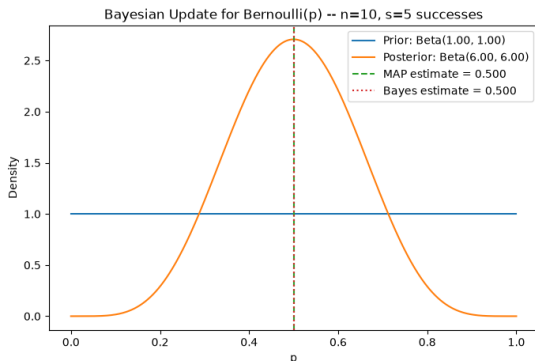
Estimates from posterior $\text{Beta}(a', b')$:

$$\hat{p}_{\text{MAP}} = \frac{a' - 1}{a' + b' - 2}, \quad \hat{p}_{\text{Bayes}} = \frac{a'}{a' + b'}$$

- **Why Beta?** Beta is the conjugate prior for Bernoulli because the posterior has the same Beta functional form as the prior – updating reduces to incrementing the shape parameters: $a' = a + s$, $b' = b + (n - s)$

Bernoulli Example: Toy Data

Prior Beta(1,1), data 1100101100 ($n = 10$, $s = 5$ successes)

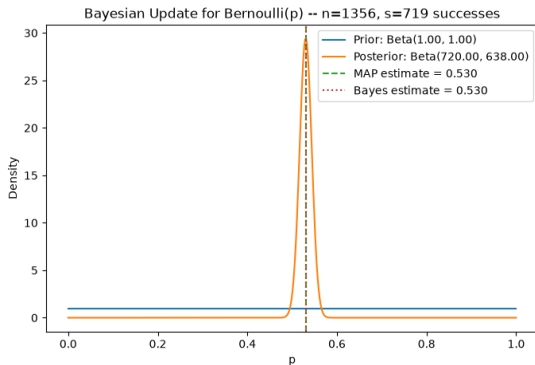


Posterior Beta(6, 6), $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} = 0.500$

- MAP and Bayes estimates coincide here because Beta(6, 6) is symmetric; for an asymmetric posterior they would generally differ

Bernoulli Example: Is AAPL's Daily Return Positive?

$X_t = 1$ if AAPL's daily log return is positive, 0 otherwise. Prior Beta(1, 1), $n = 1356$ days, $s = 719$ positive days.



Posterior Beta(720, 638), $\hat{p}_{\text{MAP}} = \hat{p}_{\text{Bayes}} \approx 0.530$

- AAPL closes up on $\approx 53\%$ of days – slightly more often than not
- 95% credible interval: [0.5036, 0.5567] – AAPL closes up slightly more than half the time, but with meaningful uncertainty remaining

Bayesian Updates: Normal Data

Model $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with **known** σ (e.g. the mean daily return μ of a stock).

Prior: $\mu \sim N(\mu_0, \sigma_0^2)$, default $\mu_0 = 0$, $\sigma_0 = 1$

Posterior (conjugacy), with sample mean $\bar{x}$:

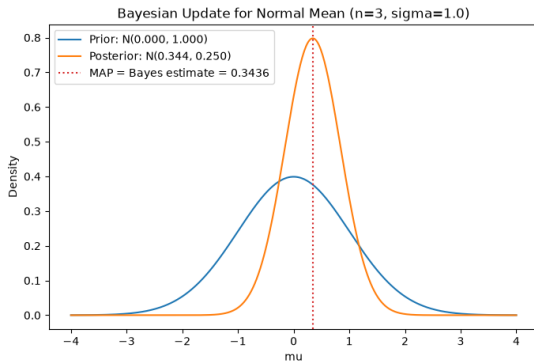
$$\mu \mid \text{data} \sim N(\mu_n, \sigma_n^2), \quad \frac{1}{\sigma_n^2} = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}, \quad \mu_n = \sigma_n^2 \left(\frac{\mu_0}{\sigma_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

For a Normal posterior, mode = mean:

$$\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = \mu_n$$

Normal Example: Toy Data

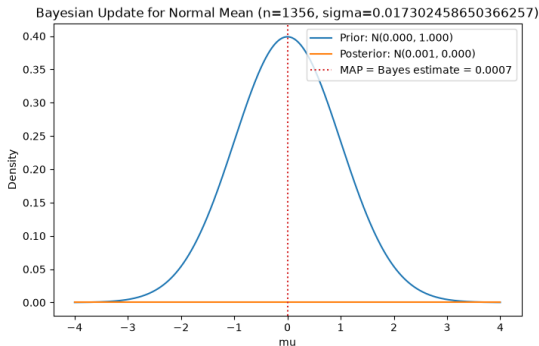
$\sigma = 1$, prior $N(0, 1)$, data 0.123, 1.3812, -0.13 ($n = 3$)



Posterior $N(0.344, 0.25)$, $\hat{\mu}_{\text{MAP}} = \hat{\mu}_{\text{Bayes}} = 0.344$

Normal Example: Mean of AAPL Daily Returns

σ fixed at the sample std of AAPL's log returns, prior $N(0, 1)$, $n = 1356$



Posterior mean $\hat{\mu}_{\text{Bayes}} \approx 0.0007$ – with $n = 1356$, the data completely dominates the (wide) prior

- 0.07% daily return compounds to approximately 19.28% annually (252 trading days) – statistically detectable but economically modest drift

Why Bayes? From Estimate to Decision

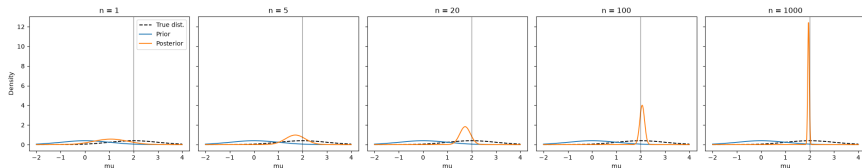
A frequentist gives a point estimate $\hat{p} = 0.530$ and a confidence interval. The Bayesian posterior gives the **full probability distribution** over p – so we can make direct probability statements:

- $P(p > 0.5 \mid \text{data}) = 0.987$ (98.7%)
- $P(p < 0.5 \mid \text{data}) = 0.013$ (1.3%)

We are 98.7% confident AAPL is an upward-drifting asset – a statement a frequentist confidence interval cannot make directly.

Supplementary: Does the Posterior Forget the Prior?

Known true $\mu = 2$, $\sigma = 1$; prior $N(0, 1)$ (deliberately off-center). Generate n points from the true distribution and compute the posterior.



- As n grows, the posterior narrows and shifts toward the true μ
- The prior's influence vanishes as $n \rightarrow \infty$
- **Bernstein–von Mises theorem:** as $n \rightarrow \infty$, the posterior converges to a Normal distribution centered at the true parameter, with variance equal to the inverse Fisher information – regardless of the prior
- **Implication:** with enough data, Bayesian and frequentist inference agree. The prior only matters when data is scarce

What the Three Methods Agree On

- **Linear Regression:** only contemporaneous information explains returns – lagged predictors have $R^2 \approx 0$
- **Bayesian Analysis:** stable long-run parameters exist ($\hat{p} \approx 0.530$, $\hat{\mu} \approx 0.0007$) but with uncertainty quantified via the full posterior
- **Markov Chains:** regime structure is persistent, but mechanically so due to the rolling-window construction
- **Common thread:** weak-form market efficiency – daily prices are unpredictable, but statistical structure exists at longer timescales

- **Tickers:** ^IXIC (NASDAQ Composite), AAPL (Apple), GOOGL (Alphabet)
- **Period:** 2021-01-01 to 2026-06-01, daily data (≈ 1356 observations)
- **Variable:** daily log returns

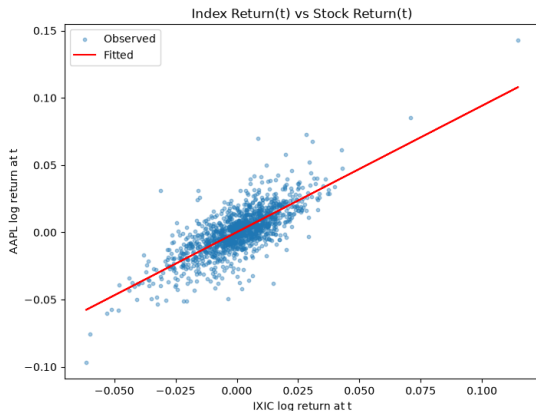
$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right)$$

- **Method:** Ordinary Least Squares (OLS), via `statsmodels`

Model 1: Index Return(t) $\rightarrow$ Stock Return(t)

Question: Can market index returns explain individual stock returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_t + \varepsilon_t$$



Model 1: Results

$$\widehat{\text{AAPL}}_t = 0.00015 + 0.940 \cdot \text{NASDAQ}_t$$

Coefficient	Estimate	<i>p</i> -value
β_0 (intercept)	0.00015	0.625 (not significant)
β_1 (NASDAQ _{<i>t</i>})	0.940	$\approx 5 \times 10^{-260}$

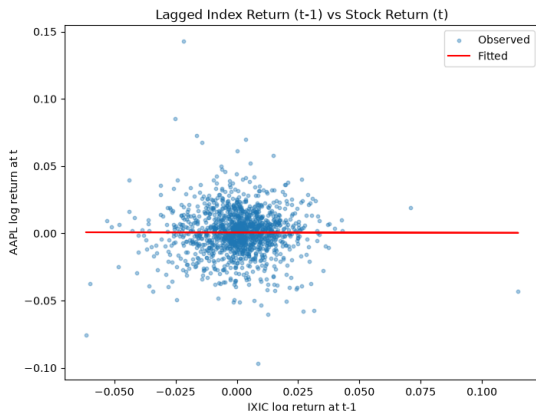
$$R^2 = 0.584, \quad \text{Adjusted } R^2 = 0.583$$

- AAPL's daily return moves almost one-for-one with the NASDAQ ($\beta_1 \approx 1$)
- $\approx 58\%$ of AAPL's return variance is explained by the index return
- Residuals are fat-tailed (kurtosis ≈ 6) – typical of daily financial returns

Model 2: Lagged Index Return($t-1$) $\rightarrow$ Stock Return(t)

Question: Can past returns help predict future returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{NASDAQ Return})_{t-1} + \varepsilon_t$$



Model 2: Results

$$\widehat{\text{AAPL}}_t = 0.00066 - 0.0024 \cdot \text{NASDAQ}_{t-1}$$

Coefficient	Estimate	<i>p</i> -value
β_0 (intercept)	0.00066	0.159 (not significant)
β_1 (NASDAQ _{<i>t</i>-1})	-0.0024	0.943 (not significant)

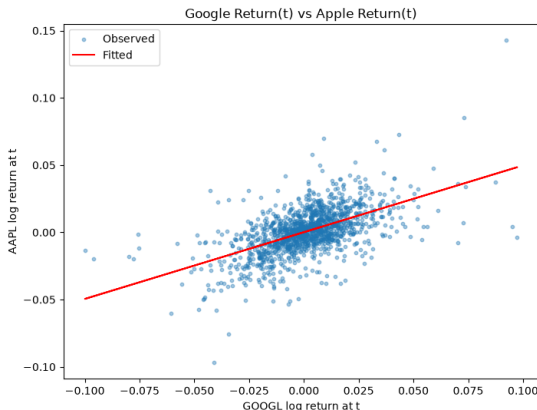
$R^2 \approx 0.0000$, Adjusted $R^2 \approx -0.0007$

- Yesterday's NASDAQ return has **no** predictive power for today's AAPL return
- Consistent with the (weak-form) Efficient Market Hypothesis

Model 3: Google Return(t) $\rightarrow$ Apple Return(t)

Question: Can one stock's returns explain another stock's returns?

$$\text{AAPL Return}_t = \beta_0 + \beta_1 \cdot (\text{GOOGL Return})_t + \varepsilon_t$$



Model 3: Results

$$\widehat{\text{AAPL}}_t = 0.00013 + 0.496 \cdot \text{GOOGL}_t$$

Coefficient	Estimate	<i>p</i> -value
β_0 (intercept)	0.00013	0.750 (not significant)
β_1 (GOOGL _{<i>t</i>})	0.496	$\approx 6.5 \times 10^{-113}$

$$R^2 = 0.314, \quad \text{Adjusted } R^2 = 0.313$$

- A 1% same-day move in GOOGL is associated with a $\approx 0.5\%$ move in AAPL
- $\approx 31\%$ of AAPL's variance explained – less than the index (Model 1, 58%)
- Suggests a common market/sector factor, not a direct causal link

Stage 1: Summary

Model	Predictor	R^2	Significant?
1	NASDAQ _t	0.584	Yes
2	NASDAQ _{t-1}	0.000	No
3	GOOGL _t	0.314	Yes

- Same-day market index return strongly explains AAPL's return ($R^2 = 0.58$)
- Lagged (past) returns carry **no** predictive information – market efficiency
- Same-day peer-stock (GOOGL) return also explains a sizeable share ($R^2 = 0.31$), likely via shared exposure to the market factor

Model A: Full Lagged Model

Question: Can past returns predict future returns?

$$\begin{aligned} AAPL_t = & \beta_0 + \beta_1 NASDAQ_{t-1} + \beta_2 AAPL_{t-1} + \\ & + \beta_3 MA_{20} + \beta_4 MA_{50} + \beta_5 MA_{200} + \\ & + \beta_6 VOL_{20} + \beta_7 VOL_{20} + \beta_8 VOL_{20} + \\ & + \beta_9 \Delta Volume_{t-1} + \beta_{10} VolRatio_{t-1} + \varepsilon_t \end{aligned}$$

$$R^2 = 0.010, \quad \text{Adjusted } R^2 = 0.001, \quad F\text{-test } p = 0.336$$

- Overall model is **not statistically significant**
- Only VolumeChg_lag1 is individually significant ($p = 0.031$)

Model B: Reduced Model

$$\text{AAPL}_t = \beta_0 + \beta_1 \cdot \text{NASDAQ}_{t-1} + \beta_2 \cdot \text{AAPL}_{t-1} + \varepsilon_t$$

Coefficient	Estimate	<i>p</i> -value
β_0 (intercept)	0.0007	0.201
β_1 (NASDAQ_lag1)	-0.0007	0.383
β_2 (AAPL_lag1)	0.0009	0.243

$$R^2 = 0.001, \quad \text{Adjusted } R^2 = -0.001, \quad F\text{-test } p = 0.506$$

- Mirrors Stage 1 Model 2: lagged index return still has zero predictive power
- Own lagged return adds nothing either

Model C: Volume-Focused Model

Question: Can trading volume explain/predict returns?

$$AAPL_t = \beta_0 + \beta_1 \cdot AAPL_{t-1} + \beta_2 \cdot \Delta Volume_{t-1} + \beta_3 \cdot VolRatio_{t-1} + \varepsilon_t$$

Coefficient	Estimate	p-value
β_0 (intercept)	0.0007	0.200
β_1 (AAPL_lag1)	0.0004	0.471
β_2 (VolumeChg_lag1)	-0.0013	0.032
β_3 (VolumeRatio_lag1)	0.0009	0.158

$R^2 = 0.005$, Adjusted $R^2 = 0.002$, F -test $p = 0.152$

- Overall model not significant, but VolumeChg_lag1 is ($p = 0.032$)
- With 10 predictors tested in Model A, one hit at $p \approx 0.03$ is roughly what chance alone would produce

Model Comparison

Model	R^2	Adj. R^2	AIC	BIC
A (full lagged)	0.010	0.001	-6053.4	-5997.9
B (reduced)	0.001	-0.001	-6059.5	-6044.3
C (volume-focused)	0.005	0.002	-6061.4	-6041.2

- Model C has the lowest AIC; Model B has the lowest BIC – the differences are negligible
- Model A is worst on both criteria (most parameters, no gain in fit)
- **None** of the three models pass the overall F -test at the 5% level

Residual Diagnostics (Model C)

- No significant autocorrelation in residuals
- Residuals are fat-tailed (kurtosis ≈ 8.7) – normality is rejected
- Same pattern as Stage 1 – typical of daily financial returns, and largely independent of which (weak) model is used

Stage 2/3: Summary

- **Market index returns:** explain AAPL *contemporaneously* ($R^2 = 0.58$, Stage 1), but the lagged index has **zero** predictive power ($R^2 \approx 0$)
- **Past returns:** across Models A, B, C, $R^2 \leq 1\%$ and the F -test is never significant – past returns do **not** predict future returns
- **Trading volume:** VolumeChg_lag1 is marginally significant alone ($p \approx 0.03$), but the overall volume model is not ($p = 0.152$) – weak, inconclusive evidence
- **Conclusion:** results are consistent with (weak-form) market efficiency – information is priced in immediately, leaving no exploitable pattern in lagged data

Defining Market Regimes

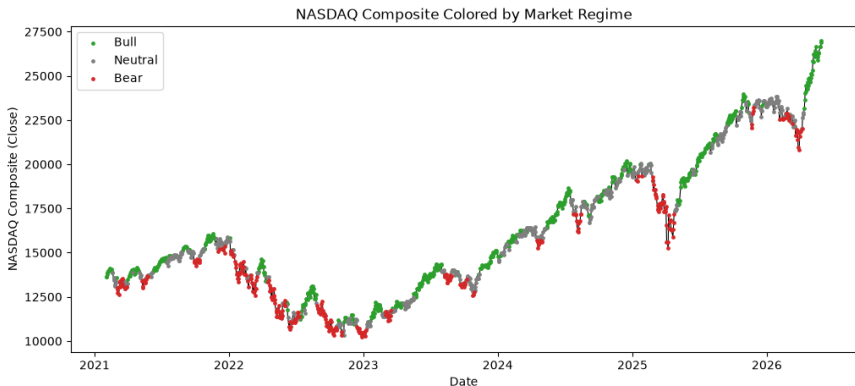
Classify each day's NASDAQ Composite (^IXIC) into **Bull**, **Neutral**, or **Bear** using the 20-day rolling cumulative log return

$$R_t = \sum_{i=0}^{19} r_{t-i}, \quad \theta = 0.5 \cdot \text{std}(R_t)$$

$$\text{Regime}_t = \begin{cases} \text{Bull} & R_t > \theta \\ \text{Bear} & R_t < -\theta \\ \text{Neutral} & \text{otherwise} \end{cases}$$

Regime	Bull	Neutral	Bear
Days (%)	554 (41.4%)	465 (34.8%)	318 (23.8%)

NASDAQ Composite Colored by Regime



Transition Probability Matrix

$$P_{ij} = P(\text{Regime}_{t+1} = j \mid \text{Regime}_t = i)$$

$$P = \begin{pmatrix} 0.879 & 0.121 & 0.000 \\ 0.144 & 0.751 & 0.105 \\ 0.000 & 0.154 & 0.846 \end{pmatrix} \quad (\text{rows/cols: Bull, Neutral, Bear})$$

- All three regimes are highly **persistent** (diagonal ≈ 0.75 – 0.88)
- $P(\text{Bull} \rightarrow \text{Bear}) = P(\text{Bear} \rightarrow \text{Bull}) = 0$ – the market always passes through Neutral

Stationary Distribution

$$\pi P = \pi, \sum_i \pi_i = 1$$

Regime	Stationary π	Observed frequency
Bull	0.414	0.414
Neutral	0.348	0.348
Bear	0.238	0.238

- Stationary distribution matches observed frequencies almost exactly – the chain is essentially already at its long-run distribution

Predicting the Next Regime

Given the current regime i , the most probable next regime is $\arg \max_j P_{ij}$.

- Last observed regime: **Bull**
- $P(\text{Bull} \rightarrow \text{Bull}) = 0.879$, $P(\text{Bull} \rightarrow \text{Neutral}) = 0.121$,
 $P(\text{Bull} \rightarrow \text{Bear}) = 0.000$
- Predicted next regime: **Bull**

Simulating the Markov Chain

Simulate a regime sequence of the same length from P and compare to the observed data.

Regime	Frequency	
	Observed	Simulated
Bull	0.414	0.476
Neutral	0.348	0.301
Bear	0.238	0.223

Regime	Avg. run length (days)	
	Observed	Simulated
Bull	8.1	8.8
Neutral	4.0	3.3
Bear	6.5	6.0

- A first-order Markov chain reproduces both the regime frequencies and the average regime durations reasonably well

- Market regimes (Bull/Neutral/Bear) are **highly persistent**, and the chain is essentially at its stationary distribution
- The model predicts the current Bull regime will most likely continue
- However, persistence is largely **mechanical**: R_t is a 20-day rolling sum, so R_t and R_{t-1} share 19 of 20 terms and are highly correlated by construction
- Consistent with Stage 1: *daily returns* themselves remain essentially unpredictable (weak-form efficiency); only the smoothed regime label is persistent

Questions?

Thank you!